
Signal & Data Analysis in Neuroscience
2020
Part 5: Multiple Point Processes

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Neural Interaction

- Event related activity
- Appendix: Real data is complicated

Suggested reading:
• Neuronal spike trains and stochastic point processes, II. Simultaneous spike trains, Perkel DH, Gerstein GL, Moore GP, Biophysical Journal 1967

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Neural interaction I

- Synaptic coupling
- Common input

SYNAPTIC COUPLING

COMMON INPUT

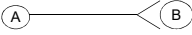
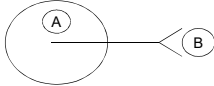
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Neural Interaction II

- Single neuron interaction
 
- Population interaction
 
- Global co-activation (arousal, task, ...)

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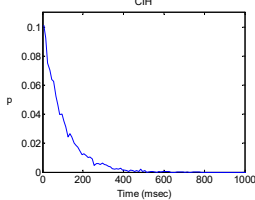





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CIH

- CIH (cross interval histogram)
- Spikes for neuron A are the renewal points for neuron B
- Exponential distribution if no interaction (assuming Poissonian neurons)



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Cross-correlation practicalities

- Count $Q_{C_{1,2}}(\tau) = \sum_{i=1}^{n_1} \rho_1(t_{1,i}) \cdot \rho_2(t_{1,i} + \tau)$
- Probability $Q_{P_{1,2}}(\tau) = \frac{1}{n_1} \cdot \sum_{i=1}^{n_1} \rho_1(t_{1,i}) \cdot \rho_2(t_{1,i} + \tau)$
- Rate $Q_{R_{1,2}}(\tau) = \frac{1}{n_1 \cdot \Delta t} \cdot \sum_{i=1}^{n_1} \rho_1(t_{1,i}) \cdot \rho_2(t_{1,i} + \tau)$

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Cross-correlation (rate normalized)

- Count

$$Q_{C_{1,2}}(\tau) = \sum_{i=1}^{n_1} [\rho_1(t_{1,i}) - r_1] \cdot [\rho_2(t_{1,i} + \tau) - r_2]$$
- Probability

$$Q_{P_{1,2}}(\tau) = \frac{1}{n} \cdot \sum_{i=1}^{n_1} [\rho_1(t_{1,i}) - r_1] \cdot [\rho_2(t_{1,i} + \tau) - r_2]$$
- Rate

$$Q_{R_{1,2}}(\tau) = \frac{1}{n \cdot \Delta t} \cdot \sum_{i=1}^{n_1} [\rho_1(t_{1,i}) - r_1] \cdot [\rho_2(t_{1,i} + \tau) - r_2]$$

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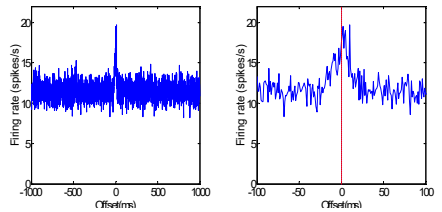
(A) Symmetry

- Reference neuron
- Target neuron
- Autocorrelation
 - $C_i(t) = C_i(-t)$
- Cross-correlation
 - $C_{ij}(t) \neq C_{ji}(t)$
 - $C_{ji}(t) = \alpha C_{ij}(-t)$

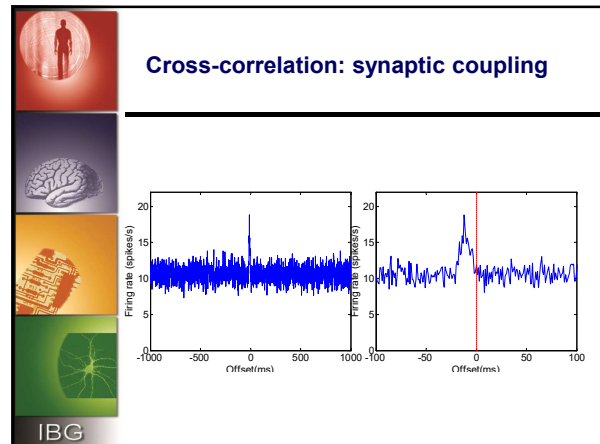
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Cross-correlation: common input



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Oscillatory correlation

- Depends on frequency of the two neurons.
- Depends on phase of the two neurons.
- Phase and frequency slips will destroy global correlations.
- Spectral analysis reveals such correlations much better.

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Summary

- Cross correlations are the most common tool for testing neural connectivity and functional relations between and within brain areas.
- Cross correlations provide a very limited ability to research the neural connectivity.
- Stronger methods exist which look at the spike sequences using measures such as mutual information.

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- Neural Interaction
- Event related activity
- Appendix: Real data is complicated

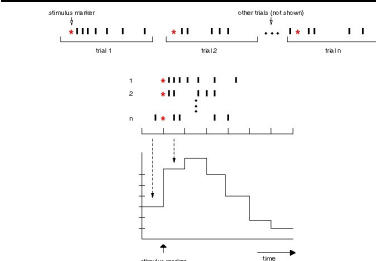
Abeles M. Quantification, smoothing, and confidence limits for single-units' histograms. J. Neuroscience Methods, 1982; 5: 317-25.

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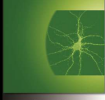

Peri-Stimulus Time Histogram (PSTH)



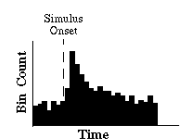
- Peri = Surrounding.
- Sometimes called **post**-stimulus time histogram when considering changes only after the event

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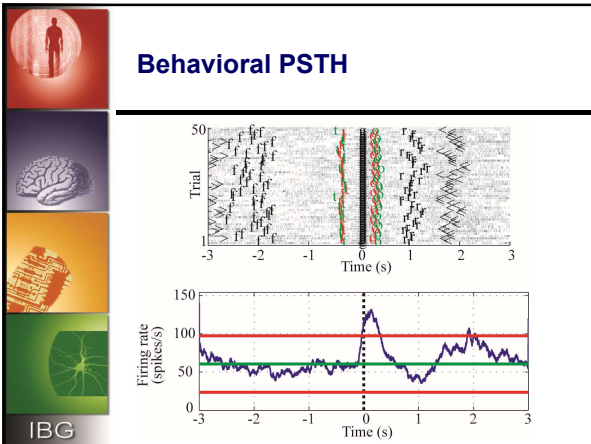



PSTH

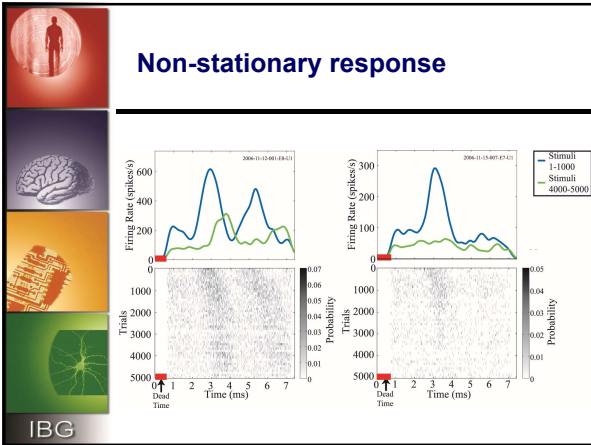


- Normalization
 - Count
 - Probability
 - Rate
- What is the determining factor:
 - Maximum response.
 - Length of the response.
 - Area (integral) of the response.

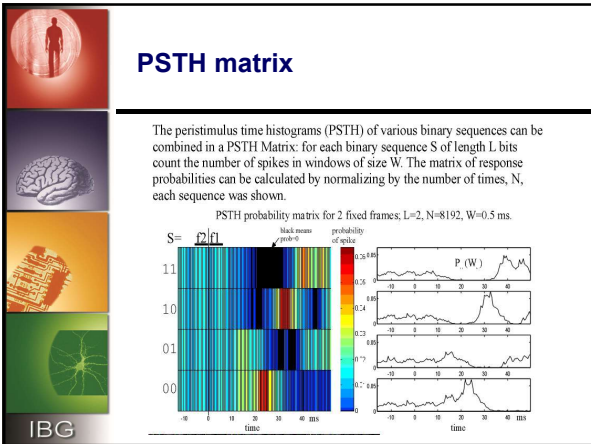
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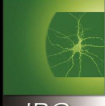


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Shift predictor I

- Remove non direct effects from the correlation.
- Example: all neurons tend to elevate their firing during activity/stimulus.
- Shift (or realign) according to trials/stimulus.
- Shift may be preformed once or on multiple moves or permutations.

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Shift predictor II

A

trial 1 trial 2

B

trial 2 trial 3

raw shift predictor corrected

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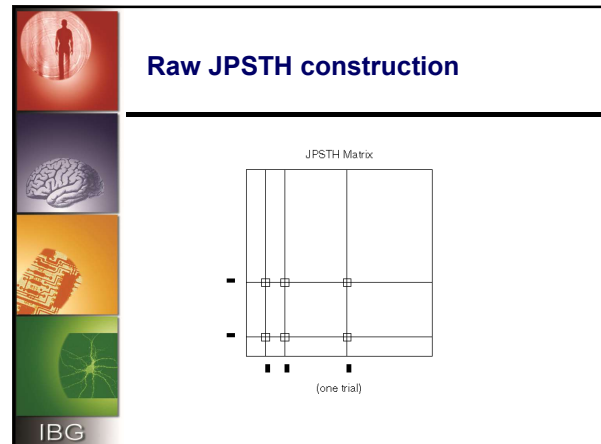



JPSTH

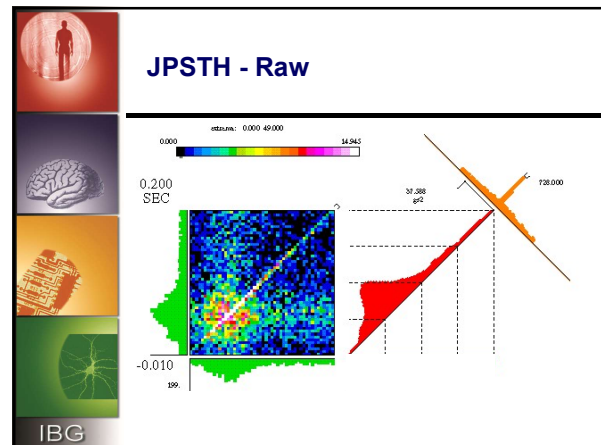
- Analysis of the dependence of correlations on the stimulus.
- Requires much more data since n^2 bins are used instead of the n bins used for PSTH or cross correlation.

JPSTH tutorial: <http://mulab.physiol.upenn.edu/jpst.html>

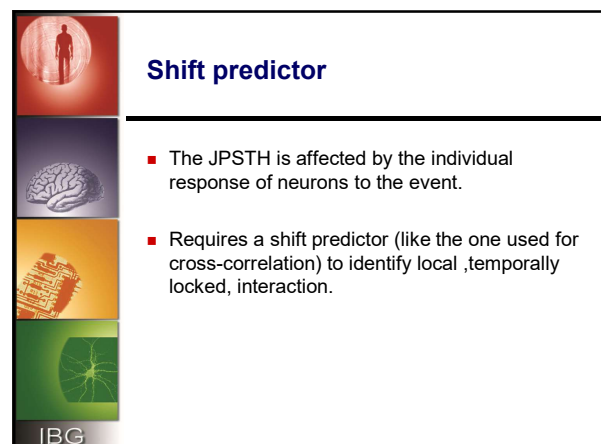
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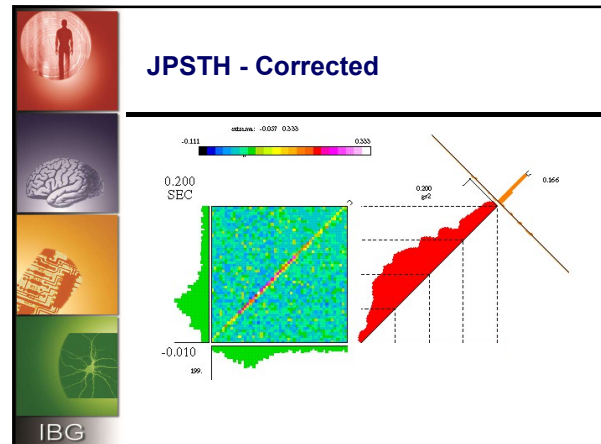
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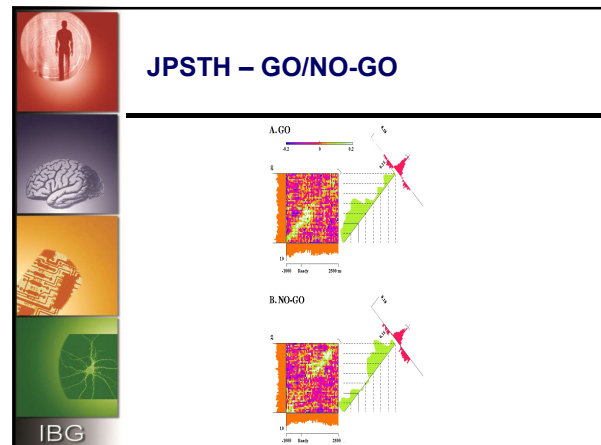
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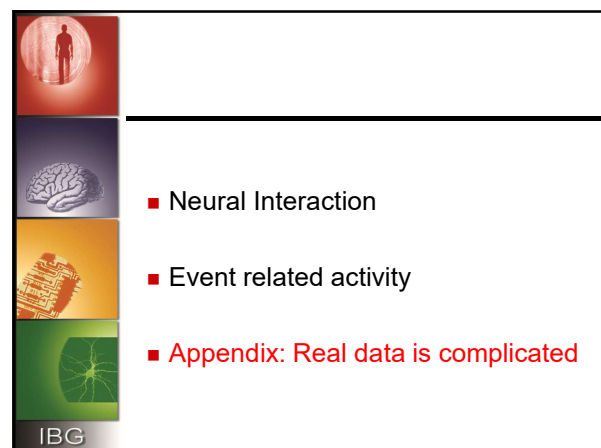
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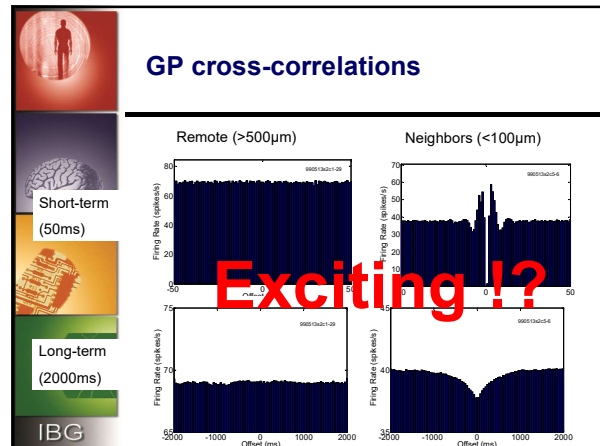


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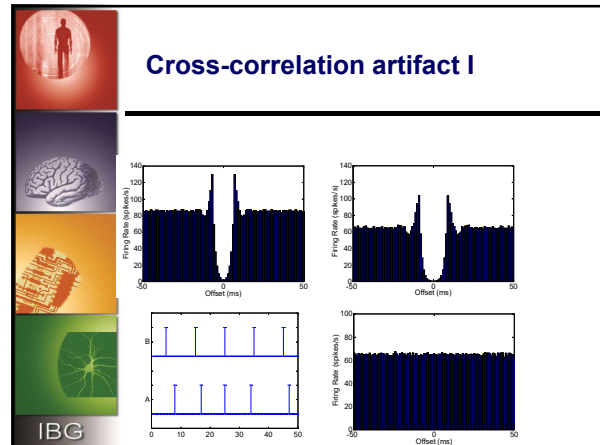


- Neural Interaction
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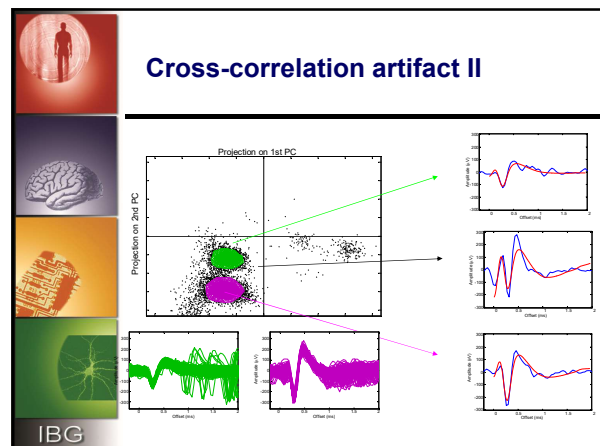
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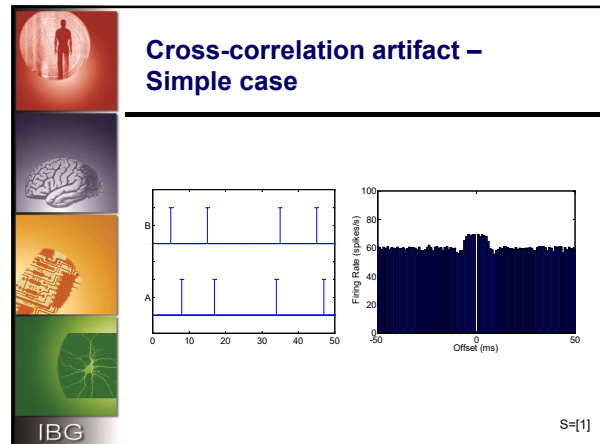
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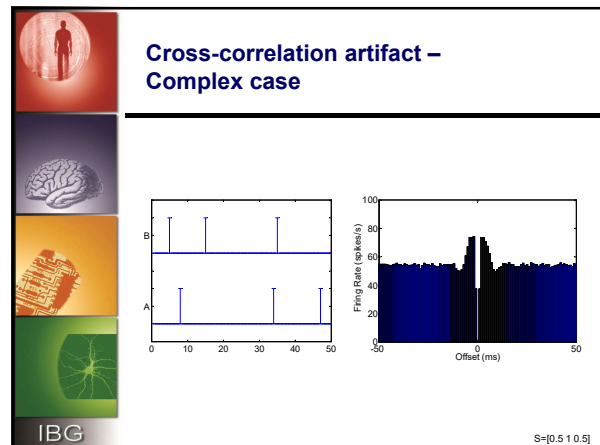
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Intuition 😊

- During the refractory period one of the cells does not fire so the other cell does not 'lose' spikes and therefore reflect the original firing rate of the cell and not the rate after removal of common spikes.
- More generally, the probability of 'losing' a spike depends on the shape of the autocorrelation functions.

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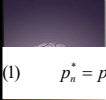
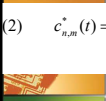

Analysis – simple case

(1) $p_n^* = p_n \cdot (1 - p_m)$

(2) $c_{n,m}^*(t) = (1 - a_n(t)) \cdot (1 - a_m(t)) \cdot \frac{p_n}{1 - p_n}$



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Analysis – complex case

(1) $p_n^* = p_n \cdot (1 - p_m \cdot \sum_{u=-\infty}^{\infty} S_{n,m}(u))$

(2) $c_{n,m}^*(t) = (1 - \sum_{u=-\infty}^{\infty} S_{n,m}(u) \cdot a_n(t+u)) \cdot (1 - \sum_{u=-\infty}^{\infty} S_{n,m}(u) \cdot a_m(t-u)) \cdot \frac{p_n}{1 - p_n \cdot \sum_{u=-\infty}^{\infty} S_{n,m}(u)}$



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Size of peak in different brain areas

	Observed Firing rate (spikes/s)	Original Firing rate (spikes/s)	$\Delta\lambda$ (Hz)	$\Delta\lambda/\lambda_{\infty}$ (%)
Pallidum	60	78.5	42.6	71.1
STN	25	27.2	4.6	18.6
Cortex	5	5.1	0.2	3.1



$S_{n,m} = S_{m,n} = [0.25 \ 0.75 \ 1 \ 0.75 \ 0.25]$

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